



# TrashUQ

*Edge AI + Federated Learning Platform*

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# Project Overview

**TrashUQ** is an end-to-end platform for intelligent waste monitoring.

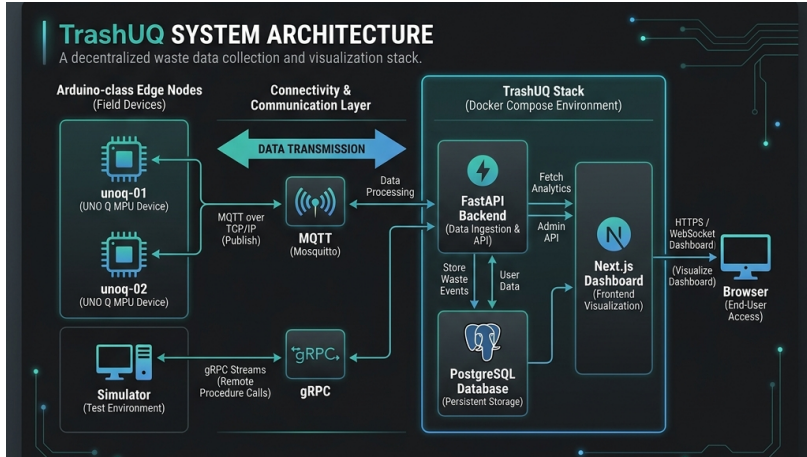
- **Decentralized Training:** Federated Learning at the Edge.
- **Real-Time Pipeline:** MQTT telemetry & WebSocket dashboard.
- **Robust Persistence:** PostgreSQL storage of every event.
- **Validated Hardware:** Deployed on Arduino-class MPU nodes.

## Core Stack

- **Backend:** FastAPI & gRPC
- **Edge:** TFLite & Python
- **Dashboard:** Next.js
- **Broker:** Mosquitto



# System Architecture



*Three-Stage Pipeline: Edge Devices → Connectivity → Deployable Stack*

## Device Layer

- **Hardware:** UNO Q MPU (Arduino-class).
- **Runtime:** TFLite-based material classification.
- **Simulator:** Full-fidelity reproducible demo mode.

## Classification Path

- Cardboard
- Glass
- Paper
- Plastic

*Telemetry: Published via MQTT topics.*

# The Communication Backbone

## MQTT Telemetry

- Real-time status, metrics, and classification streams.
- Low-bandwidth, high-reliability pub/sub.

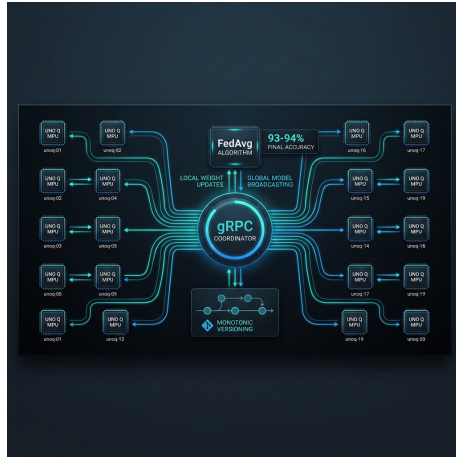
## gRPC Coordination

- High-performance binary RPC for model weight transfer.
- Concurrency-safe: Rejects stale updates during rounds.

### Data Flow

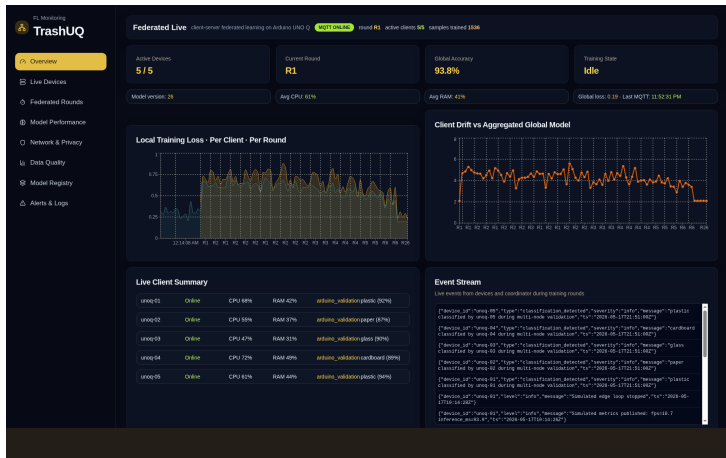
arduino/+/# → Backend →  
PostgreSQL → Dashboard

# Federated Learning Scaling



*FedAvg Validation: Scalability study across 2, 5, 10, and 20 clients.*

# Next.js Live Dashboard



*Bootstrap + MQTT WebSockets: Real-time charts and device summaries.*



## Part A: Deployed Validation

- **Infrastructure:** 2 nodes (unoq-01, unoq-02) on bepes-server.
- **Persistence:** 282 MQTT messages stored with full fidelity.
- **Monotonicity:** Advanced `model_version` from 0 to 26 over 26 rounds.
- **Reliability:** 0% invalid payloads, 0% dropped messages.

| Node    | Messages | Mean Accuracy |
|---------|----------|---------------|
| unoq-01 | 147      | 92.1%         |
| unoq-02 | 135      | 87.1%         |



## Part B: Scaling Performance

- **Experiment:** FedAvg scaling study (2 to 20 clients).
- **Condition:** Non-IID Dirichlet partitioning ( $\alpha = 0.3$ ).
- **Convergence:** Stable convergence with **93-94% final accuracy**.
- **Efficiency:** predictable linear growth in communication cost.

**Conclusion:** System maintains high performance at scale under heterogeneous data.

## TrashUQ Success

- Full edge-to-cloud integration.
- Validated FL coordinator.
- Real-time dashboard.

## The Team

- Aleix Bertran
- Aleix Rosinach
- Bru Pallàs
- Pol Llinàs

# Thank You!